

Problems register — from the 35-case engine-vs-reality validation (16-Jul)

Source: `engine-vs-reality-16jul.md` (35 real FC↔final-bill pairs). This register exists so every fix is designed **not to create ten new failures**: each problem carries its root cause, the safe fix, what the naive fix would break, and the regression gate that must stay green before the fix ships.

The guardrail infrastructure (applies to every fix below)

1. **The 35-case set is now a frozen regression suite.** Extractions + verdicts live in `engine-vs-reality-16jul-data/`; the pipeline is re-runnable (`extract.mjs` skip-done, `run_engine.mjs` , `compare.py`). Rule: a fix ships only if the 12 ENGINE_GOOD cases stay within $\pm 25\%$ and no case's verdict worsens. The fix's own target cases must improve — measured, not assumed.
 2. **Existing suites that must stay green:** cash parity 1042/1042 (TKR figure is pinned to the paisa), flow2 374-case sweep (zero data-invariant failures), verify-estimates harness (207 zero-input builds).
 3. **Additive-only response policy:** no fix changes the meaning of an existing field (`final_estimate` stays itemized). New behavior arrives as new fields with a `source` /flag so the UI and the audits can attribute every number.
 4. **One flag per behavior change**, so a bad fix can be turned off without a rollback of unrelated work.
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P1 — Package route computes the right total but never quotes it

Cases/money: SURLA +82% (₹330k phantom error), G NAGAVENI (no usable number either way); touches every package-billed quote — 8/35 real bills were packages. **Why:** `final_estimate` is always the itemized figure. When billing identification says "package", `package_offer` carries the package and even the conversion machinery — but no `with_package` headline ever reaches the quote. The engine's own arithmetic (pkg 224,600 + ~190k implant excludes) reproduces SURLA's ₹406,505 bill almost exactly. *The engine knows the route; it doesn't finish the sentence.* **Safe fix:** `package_offer.quote = { with_package_total, basis }` computed from package amount + predicted payable extras (the `bucket_extras` history built 16-Jul is exactly the "predicted excludes" source). Surfaces as the headline **only on the package route AND only when** readiness is `can_generate_estimate` AND the conversion check does not flag the band — otherwise the itemized figure stays headline with the package quote as secondary. Never touch `final_estimate`. **Naive fix would break:** replacing `final_estimate` with the package total → every parity pin, saved-estimate payload, PDF, and settlement calculation shifts; quoting packages with ₹10 placeholder amounts or failed conversion checks → confident garbage (G NAGAVENI's ₹27,500 tendon package is exactly that trap — its `not_ready` status must keep blocking). **Regression gate:** parity suite unchanged (cash TKR is itemized-route); 35-case replay — SURLA/GUDURU/AKARAPU improve, HEMALATHA/PINNAMANENI stay exact; verify harness unchanged (additive field).

P2 — "NON ROBOTIC" wording triggers the ₹230,000 robotic add-on

Cases/money: SURLA (+₹230k on one case; any negated-wording GIPSA TKR). **Why:** the `treatment_text_robotic` detector is a substring match (`/ROBOT/i`) — "TKR **NON ROBOTIC** B/L" contains "ROBOT". Verified: `robotic:"no"` rebuild drops 737,751 → 507,751. **Safe fix:** negation guard in ONE place (the wording detector used by both build and flow2): treat as robotic only when the robot token is NOT immediately preceded by a negator (`NON[- ]?` , `NOT` , `NO` , `WITHOUT`). Nothing else in the robotic ladder changes — presence rules, gate flag, explicit `robotic:"yes"/"no"` all keep their current precedence. **Naive fix would break:** a broad "any negation word anywhere in the text" rule would suppress genuine robotic on wordings like "robotic TKR, non-weight-bearing post-op". Anchor the negator to the token, not the sentence. **Regression gate:** flow2 sweep robotic questions unchanged; the robotic matrix repro (tmp-repro-robotic-matrix) — explicit robotic wordings still include ₹1.2L/₹2.3L; 35-case replay — SURLA improves, HEMALATHA/PINNAMANENI/ LAKSHMI (genuinely robotic) unchanged.

P3 — Named high-cost drugs invisible to daycare infusion pricing

Cases/money: SHOURYA -87% (₹119k missed), VARSHITHA -64% (₹32k). Humans were within 8–13% — they simply looked the drug up. **Why:** the infusion family prices the generic cohort; the drug name sits in the treatment text AND in the pharmacy master (MRP lookup exists), but no path connects them. **Safe fix:** for daycare-infusion-class families ONLY (explicit family whitelist), attempt a high-confidence pharmacy-master match on the treatment text; on a hit, add a `named_drug_line` (MRP × qty) and **replace** the cohort pharmacy figure with `max(cohort_pharmacy_P50, drug_line + non-drug_pharmacy_residual)` — replace, not add, so the drug isn't double-counted where the cohort already partially carries it. Line flagged `source: named_drug_mrp` + a warning asking the FC to confirm the drug/qty. **Naive fix would break:** plain addition double-counts (cohort pharmacy P50 already includes some drug spend); running the text-match on ALL families would bolt drug lines onto surgical estimates from stray brand names in remarks. The whitelist + replace-not-add + visible flag prevent all three. **Regression gate:** 35-case replay — SHOURYA/VARSHITHA improve; every non-infusion case byte-identical (whitelist guarantees it); chemo/immuno families in the verify harness re-checked for double-count.

P4 — Catch-all families match specific wording with unearned confidence

Cases/money: BANDANADAM -74% (₹332k under), KRISHNA +371%, G NAGAVENI -36%. **Why:** when the wording names something we have no family for (maxillofacial arch-bar, complex plastic reconstruction, Achilles repair), the AI matcher falls to a catch-all (general_plastic, orthopaedic_management, ...) and reports medium/high confidence. The estimate inherits a cohort that has nothing to do with the case. **Safe fix — the distinction that prevents collateral:** generic wording → generic family is CORRECT ("medical management" → general_medical_management scored -15%/-26% GOOD and must not regress). The failure is **specific wording** → **generic family**. Guard: top match ∈ catch-all list AND the wording contains specific procedure tokens with no overlap against that family's label/aliases ⇒ cap confidence at `low` + set `needs_confirmation: true`; flow2 renders its existing pending-question machinery ("we only have a generic match — confirm or pick"), the Simple flow shows its existing confirm step with a visible generic-match warning. No new blocking anywhere else; explicit user confirm proceeds exactly as today. **Naive fix would break:** demoting catch-all's wholesale → NO_MATCH storms on the (correct) generic-wording majority; asking a question on every catch-all → question fatigue and a broken 374-sweep. The specific-tokens-without-overlap condition is what scopes it. **Regression gate:** 35-case replay — ARTH/NEEHARIKA/Baby-of-NIKHAT (generic wording, GOOD) still auto-resolve; BANDANADAM/KRISHNA/NAGAVENI now stop at a question (counted as converted-to-question, not as a miss); flow2 sweep re-run — no new pending questions on families whose label matched the wording.

P5 — Neonatal coverage gap (and the humans share it)

Cases/money: 3 "Baby of..." cases +79% to +143% (quotes ₹63k vs bills ₹26–36k). Human FCs were +89% to +169% — a mined newborn family beats both. **Why:** no newborn/nursery family exists; "Baby of X, medical management" resolves confidently to ADULT general_medical_management, whose cohort floor is adult-priced. **Safe fix:** data work first — mine a `newborn_care_nursery` family from the billed corpus (the FB set itself shows the ₹26–36k @1–3d shape). Routing: "Baby of"/newborn/neonate token OR age < 30 days ⇒ prefer the newborn family. Age check prevents the 10-year-old ENT case from routing to nursery. **Naive fix would break:** keyword-only routing ("baby") without the age check grabs paediatric surgical cases (JIVITESH is "Master", age 10 — verify the token list against all 35). NICU escalations must not be forced into the nursery cohort — keep NICU wording routing to its own path. **Regression gate:** 35-case replay — the 3 neonates land near ₹26–36k; JIVITESH/CHARVIN (paediatric surgical) unchanged; flow2 sweep unchanged (new family = new registry entry, additive).

P6 — Trivial-stay floor on medical management

Cases/money: NARESH +421% (₹41k quote vs ₹7.9k bill; small money, terrible optics). **Why:** template rows backfilled from historical P50s (Investigations ₹20,790 etc.) are stay-independent — a 1-day observation inherits the full cohort's median diagnostics load. **Safe fix:** LOS-conditioned residuals for medical-management families: when the requested LOS sits at/below the cohort's P25 stay, compute backfilled bucket residuals from the SAME-stay-band sub-cohort (enough cases

exist: 1-day medical admissions are common); fall back to today's behavior when the sub-cohort is thin (<15 cases). Never applies to surgical families (their costs are procedure-driven, not stay-driven). **Naive fix would break:** scaling buckets linearly by LOS ratio would gut correct 2–3 day estimates (ARTH -14% replay, NEEHARIKA -15% must not move) and mis-model fixed per-admission costs. Sub-cohort quartiles, not scaling. **Regression gate:** 35-case replay — NARESH shrinks toward ₹8–15k; ARTH/NEEHARIKA/SARBESWAR-replay unchanged (their LOS $\geq$ cohort P25); verify harness medical families re-run (this touches the same residual machinery as 15-Jul Q3 — its report must not regress).

P7 — Package-master amounts lag the billed tariff (data drift)

Cases/money: ORT5510 master 149,900 vs billed 168,700; ORT5531 master 224,600 vs billed 202,100 AND 252,600 (two amounts, same code — likely tariff version/room-tier variants). ₹19–28k per package quote. **Why:** the master was loaded once; the hospital's contracted amounts moved. The engine's own conversion-check already fires on exactly these packages — the knowledge exists but reads as a footnote. **Safe fix:** this IS the already-queued package-tariff sync audit (manager's own design, 15-Jul): billed package data as source of truth, per tariff code $\times$ room type, documenting nominal-vs-significant variances — then a reviewed master refresh. Until then: promote the conversion-check warning to quote-level prominence on the package route (ties into P1's gating), never silently. **Naive fix would break:** bulk-overwriting master amounts from bills conflates room-tier price variants (the two ORT5531 amounts are probably both correct for different tiers) and destroys the audit trail. Audit $\rightarrow$ review $\rightarrow$ refresh. **Regression gate:** P1's gate covers the display side; the sync audit itself produces the diff sheet before any master row changes.

P8 — Combo quoting (urology-class) — known, scoped, not a regression risk

Cases/money: RAMBABU -68% even at actual LOS; 5/35 bills were 50%/25% multi-procedure combos. ENT combos are already absorbed by cohort history (+13%/-8%) — fixing combos must not disturb that. **Why:** the engine prices one treatment; the hospital's combo ladder is mechanical (pkg + 2nd procedure @50% + others @25%) but unmodelled. Detection

- per-path evaluation shipped 16-Jul (flow2 + intake announcement); combined pricing was explicitly deferred. **Safe fix (when taken up):** a `combo_quote` composed from flow2's per-path package identifications through the 50/25 ladder, offered as a labeled secondary quote — never replacing single-treatment pricing, and NOT applied to ENT-class families whose cohorts already contain combo bills (that would double-count; the family's own billed history tells us which class it is: compare cohort P50 vs single-procedure package price). **Regression gate:** JIVITESH/MEENAKSHI (ENT combos, already good itemized) must not move; RAMBABU-class gets the new secondary quote.

P9 — Recounselling scored as initial estimates

Cases: 7/35 FCs were recounsellings; 4 ended course-changed. **Why:** a mid-treatment estimate needs consumed-to-date context the engine never receives; comparing it to the full final bill misjudges everyone. **Safe fix:** not an engine change — a scoring/product note: recounselling quotes should eventually take "billed so far" as an input (future feature); until then the regression suite tags them (R) and weighs them accordingly.

Watch-list (not fixed deliberately)

- RAMULU +27%** (Investigations bucket ₹55,490 looks inflated) — single marginal case; folded into the routine bucket audit rather than a targeted change that could nudge other ortho cases.
- Y CHANDRIKA bilateral EVLT -33%** — laterality/consumable modeling; needs bilateral sub-cohorts. Deferred: low case volume, medium complexity, and a wrong generic "x2" rule would damage unilateral estimates.
- KRANTHI FB extraction inconsistency** (category sum $\neq$ gross) — data-entry or extraction issue on the bill side; excluded from scoring, flagged.

Order of execution (money $\times$ safety)

P2 (one-line, zero-risk) → P1 (biggest money, additive) → P3 (whitelisted) → P4 (converts misses to questions) → P5 (data mining) → P6 (sub-cohort work) → P7 (the queued sync audit) → P8/P9 (scoped phases).